Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2023-2024

Comprehensive Examination
(EC-3 Makeup)

Course No.	: DSECLZG522
Course Title	: Big Data Systems
Nature of Exam	: Open Book
Weightage	: 40%
Duration	: 2.5 Hours
Date of Exam	: 21-04-2024 (FN)

No. of Pages = 2
No. of Questions = 8

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q.1. An application is deployed on a cluster with little availability support. A node in a cluster fails every 100 hours while other parts never fail. On failure of the node, the whole cluster needs to be shutdown, the faulty node is to be replaced and the cluster system to be restarted. This takes 2 hours. The application also needs to be restarted, which takes 2 hours. What is the availability of the application? If downtime cost is $40000 per hour, what is the yearly downtime cost?

[4 Marks]

Q.2. You have a 928 MB file stored on HDFS as part of a Hadoop 2.x distribution. A data analytics program uses this file and runs in parallel across the cluster nodes.
(a) The default block size and replication factor are used in the configuration. What will be the total number of blocks including the replicas that will be stored in the cluster? What are the unique HDFS block sizes you will find for the specific file?
(b) The cluster has 64 cores to speed up the processing. If the program can at best achieve 60% parallelism in the code to exploit the multiple cores and the rest of it is sequential, what is the theoretical limit on speed-up you can expect with 64 cores compared to a sequential version of the same program running on one core with the same file? How will this limit change if you doubled the compute power to 128 cores. You can simplify the system to assume cluster nodes and cores mean the same and we can ignore the overheads of communication etc. depending on the specific cluster configuration, scheduling etc.
(c) Suppose you could use a more scalable algorithm with 80% parallelism and the size of the file is doubled as you move to the 128 cores cluster. What would be the theoretical speed-up limit for the 128 cores cluster?

[6 Marks]

Q.3. What are the scenarios in which Flume and Sqoop are used? What are the major differences between Flume and Sqoop? Configure a Flume agent to transfer files from a local folder to an HDFS folder.

[6 Marks]

Q.4. What is meant by Consistency in CAP theorem? Let's establish a few definitions:

N = The number of nodes that store replicas of the data.

W = Number of replicas that need to acknowledge the receipt of update before update completes.

R = Number of replicas that are contacted when a data object is accessed through a read operation.

What type of consistency can be guaranteed in the following scenarios?

(a) $W+R > N$

(b) $N=2$, $W=2$, and $R=1$

(c) $N=2$, $W=1$, and $R=1$

[3 Marks]

Q.5. In the Spark Scala code shown below, list down the Transformations and Actions. What is the output of this code?

```
val linesDS = sc.parallelize(Seq("Spark is fast","Spark has Dataset","Spark Dataset is typesafe")).toDS()
val Nolines = linesDS.count()
val wordsDS=linesDS.flatMap(_.toLowerCase.split(" ")).filter(_ != "")
val groupedDS=wordsDS.groupBy("value")
val countsDS=groupedDS.count()
countsDS.show()
```

[5 Marks]

Q.6. You are developing a bigdata analytics application in PySpark using SparkSQL queries. The structured data file with the schema defined in the header is stored on HDFS in the form of a .CSV file. What are the different methods available for making the contents of the file available in an SQL table in your PySpark program for firing sparkSQL queries? What are the advantages and disadvantages of each of these methods?

[5 Marks]

Q.7. What is the difference between .cache() and .persist() APIs of RDDs? What are the supported storage levels for RDDs? How do you recommend the right storage level for RDDs in your Spark application?

[6 Marks]

Q.8. An eCommerce company is planning to launch a new application on cloud for online sale of items. During normal seasons of the year, this application deployed on a cluster of 4 servers will guarantee acceptable response to the customers. During festival seasons, the sales are expected to double. Which feature of cloud will be useful in minimizing the data center costs without compromising the response time. What are the recommended features of the component products to be used in the design of this application?

[5 Marks]

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